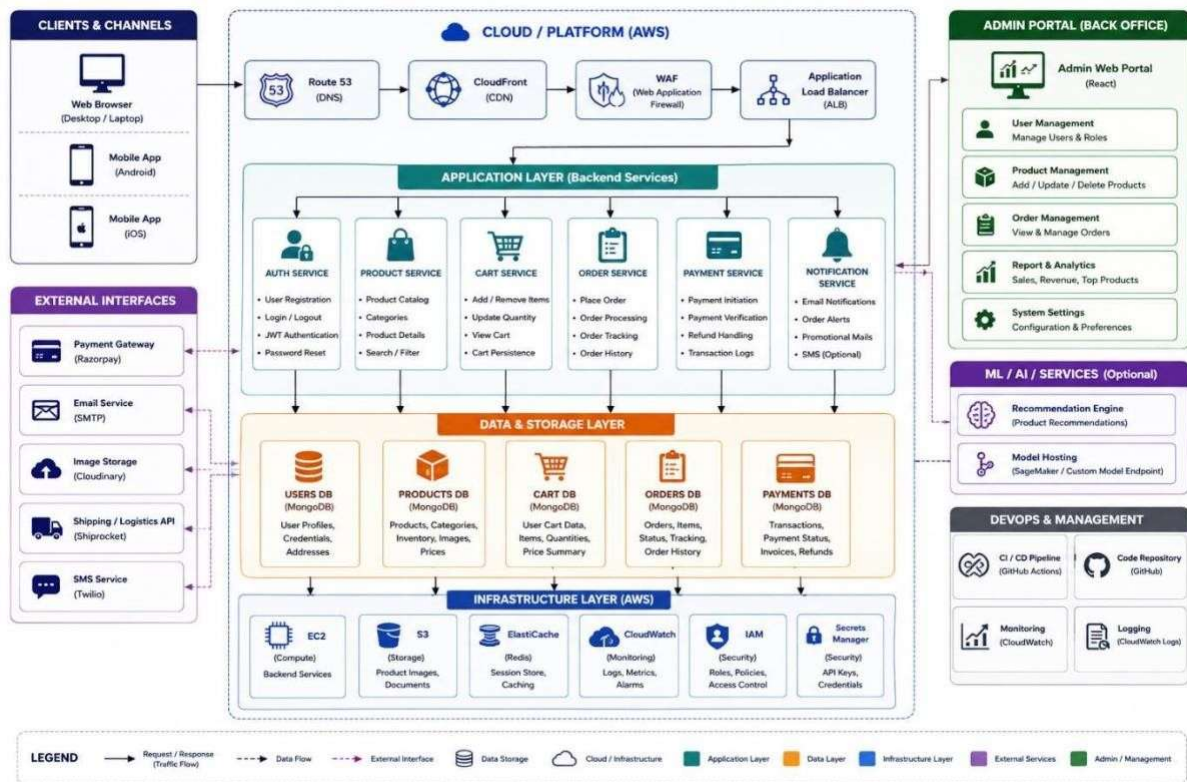


**Project Design Phase-II**  
**Technology Stack (Architecture & Stack)**

|               |                                |
|---------------|--------------------------------|
| Date          | 18 June 2026                   |
| Prepared By   | JAMPA KAILASH JOHNITH          |
| Project Name  | ShopEZ: E-Commerce Application |
| Maximum Marks | 4 Marks                        |

**Technical Architecture for ShopEZ:**



**Table-1 : Components & Technologies:**

| S.No | Component           | Description                                               | Technology                         |
|------|---------------------|-----------------------------------------------------------|------------------------------------|
| 1.   | User Interface      | Enables users to browse products and place orders.        | React.js, Vite, Tailwind CSS       |
| 2.   | Application Logic-1 | Processes user authentication, cart, and order functions. | Node.js, Express.js, JWT, bcryptjs |

|     |                                 |                                                    |                                       |
|-----|---------------------------------|----------------------------------------------------|---------------------------------------|
| 3.  | Application Logic-2             | Handles product creation, updates, and retrieval.  | Node.js, Express.js, Mongoose         |
| 4.  | Application Logic-3             | Provides tools for managing products and orders.   | Node.js, Express.js, MVC Architecture |
| 5.  | Database                        | Stores user, product, cart, and order information. | MongoDB (Mongoose ODM)                |
| 6.  | Cloud Database                  | Maintains application data on the cloud.           | MongoDB Atlas                         |
| 7.  | File Storage                    | Stores and serves product images.                  | URL-based Image Storage               |
| 8.  | External API-1                  | No external services integrated.                   | N/A                                   |
| 9.  | External API-2                  | No third-party APIs utilized.                      | N/A                                   |
| 10. | Machine Learning Model          | Not required for this application.                 | N/A                                   |
| 11. | Infrastructure (Server / Cloud) | Supports deployment and backend operations.        | Node.js Server, MongoDB Atlas         |

**Table-2: Application Characteristics:**

| S.No | Characteristics | Description | Technology |
|------|-----------------|-------------|------------|
|------|-----------------|-------------|------------|

|    |                          |                                                         |                                |
|----|--------------------------|---------------------------------------------------------|--------------------------------|
| 1. | Open-Source Frameworks   | Developed using open-source web technologies.           | React.js, Node.js, Express.js  |
| 2. | Security Implementations | Protects user accounts and application data.            | JWT Authentication, bcrypt     |
| 3. | Scalable Architecture    | Separates frontend, backend, and database layers.       | React.js, Express.js, MongoDB  |
| 4. | Availability             | Uses cloud services for reliable access.                | MongoDB Atlas, Node.js         |
| 5. | Performance              | Ensures smooth communication between system components. | MongoDB, Express.js, REST APIs |